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**IMPACTO DA OBESIDADE SARCOPÊNICA SOBRE O ESTRESSE
OXIDATIVO, OS FATORES INFLAMATÓRIOS E A CAPACIDADE
FUNCIONAL EM IDOSOS**

FRANCISCO BELTRÃO – PR
(FEVEREIRO/2026)

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**IMPACTO DA OBESIDADE SARCOPÊNICA NO PERFIL INFLAMATÓRIO E
CAPACIDADE FUNCIONAL EM IDOSOS**

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Essa dissertação foi julgada adequada para obtenção do título de Mestre em Ciências Aplicadas à Saúde e aprovada em sua forma final pelo(a) Orientador(a) e pela Banca Examinadora.

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IMPACTO DA OBESIDADE SARCOPÊNICA SOBRE O ESTRESSE OXIDATIVO, OS FATORES INFLAMATÓRIOS E A CAPACIDADE FUNCIONAL EM IDOSOS

Introdução: A obesidade sarcopênica, caracterizada pela coexistência de excesso de adiposidade e redução da massa e força muscular, tem sido associada a comprometimentos metabólicos, inflamatórios e funcionais em idosos. No entanto, os dados sobre sua prevalência e biomarcadores relacionados em populações idosas fisicamente ativas inscritas em programas de promoção da saúde ainda são limitados. **Objetivos:** Analisar a associação entre obesidade sarcopênica, marcadores inflamatórios, perfil oxidativo e capacidade funcional em idosos participantes do projeto de extensão Universidade Aberta à Terceira Idade (UNATI). **Métodologia:** Este estudo transversal incluiu 119 idosos (≥ 55 anos) inscritos em um Programa Universitário para Idosos no sul do Brasil (84,9% mulheres; idade média: $68,7 \pm 6,8$ anos). A composição corporal foi avaliada por bioimpedância elétrica, a força muscular por dinamometria de preensão manual e o desempenho físico pela velocidade da marcha. A sarcopenia foi definida de acordo com os critérios do EWGSOP2 e a obesidade sarcopênica foi identificada utilizando diferentes combinações de parâmetros de adiposidade e massa muscular. Foram analisados marcadores metabólicos, inflamatórios e de estresse oxidativo. As associações foram avaliadas usando testes bivariados e correlações de Spearman ($p \leq 0,05$). **Resultados:** A prevalência geral de obesidade sarcopênica foi de 4,2% ($n = 5$), variando de acordo com os critérios diagnósticos aplicados. As mulheres apresentaram maior prevalência de baixa massa muscular esquelética (65,3% vs. 35,3%) e força de preensão manual reduzida, enquanto os homens apresentaram maior risco de percentual de gordura (47,1% vs. 17,8%). A circunferência da panturrilha foi a única variável significativamente associada à obesidade ($p = 0,021$). Entre as mulheres, os indicadores de adiposidade correlacionaram-se positivamente com os níveis de triglicerídeos (IMC: $r = 0,263$, $p < 0,001$; massa gorda: $r = 0,295$, $p < 0,001$), enquanto a velocidade da marcha associou-se inversamente à peroxidação lipídica medida por TBARS ($r = -0,295$, $p < 0,001$), indicando menor dano oxidativo com melhor desempenho funcional. Não foram observadas diferenças significativas entre os sexos nos marcadores glicêmicos, lipídicos, inflamatórios ou de estresse oxidativo ($p > 0,05$). **Conclusões:** Embora a prevalência de obesidade sarcopênica tenha sido baixa

nesta população idosa fisicamente ativa, observaram-se associações relevantes entre adiposidade, perfil metabólico, estresse oxidativo e desempenho funcional. Esses achados destacam a importância da triagem precoce da composição corporal, força muscular e desequilíbrio redox, mesmo em ambientes que promovem a saúde, para prevenir o declínio funcional subclínico durante o envelhecimento.

Palavras-chave: Obesidade Sarcopênica, Estresse Oxidativo, Inflamação, Idoso, Capacidade Funcional.

IMPACT OF SARCOPENIC OBESITY ON OXIDATIVE STRESS, INFLAMMATORY FACTORS, AND FUNCTIONAL CAPACITY IN OLDER ADULTS

Introduction: Sarcopenic obesity, characterized by the coexistence of excess adiposity and reduced muscle mass and strength, has been associated with metabolic, inflammatory, and functional impairments in older adults. However, data on its prevalence and related biomarkers in physically active older populations enrolled in health promotion programs are still limited. Objectives: To analyze the association between sarcopenic obesity, inflammatory markers, oxidative profile, and functional capacity in older adults participating in the extension project Universidade Aberta à Terceira Idade (UNATI). Methodology: This cross-sectional study included 119 older adults (≥ 55 years) enrolled in a University Program for Older Adults in southern Brazil (84.9% women; mean age: 68.7 ± 6.8 years). Body composition was assessed by electrical bioimpedance, muscle strength by handgrip dynamometry, and physical performance by gait speed. Sarcopenia was defined according to EWGSOP2 criteria, and sarcopenic obesity was identified using different combinations of adiposity and muscle mass parameters. Metabolic, inflammatory, and oxidative stress markers were analyzed. Associations were evaluated using bivariate tests and Spearman correlations ($p \leq 0.05$). Results: The overall prevalence of sarcopenic obesity was 4.2% ($n = 5$), varying according to the applied diagnostic criteria. Women showed a higher prevalence of low skeletal muscle mass (65.3% vs. 35.3%) and reduced handgrip strength, while men had a higher risk of body fat percentage (47.1% vs. 17.8%). Calf circumference was the only variable significantly associated with obesity ($p = 0.021$). Among women, adiposity indicators were positively correlated with triglyceride levels (BMI: $r = 0.263$, $p < 0.001$; fat mass: $r = 0.295$, $p < 0.001$), while gait speed was inversely associated with lipid peroxidation measured by TBARS ($r = -0.295$, $p < 0.001$), indicating lower oxidative damage with better functional performance. No significant differences were observed between sexes in glycemic, lipid, inflammatory, or oxidative stress markers ($p > 0.05$). Conclusions: Although the prevalence of sarcopenic obesity was low in this physically active older population, relevant associations were observed between adiposity, metabolic profile, oxidative stress, and functional performance. These findings highlight the importance of early

screening of body composition, muscle strength, and redox imbalance, even in health-promoting environments, to prevent subclinical functional decline during aging.

Keywords: Sarcopenic Obesity, Oxidative Stress, Inflammation, Older Adult, Functional Capacity.

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IMPACT OF SARCOPENIC OBESITY ON OXIDATIVE STRESS, INFLAMMATORY FACTORS, AND FUNCTIONAL CAPACITY IN OLDER ADULTS

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IMPACT OF SARCOPENIC OBESITY ON OXIDATIVE STRESS, INFLAMMATORY FACTORS, AND FUNCTIONAL CAPACITY IN OLDER ADULTS

Background: Sarcopenic obesity, characterized by the coexistence of excess adiposity and reduced muscle mass and strength, has been associated with metabolic, inflammatory, and functional impairments in older adults. However, data on its prevalence and related biomarkers in physically active older populations enrolled in health promotion programs remain limited. **Methods:** This cross-sectional study included 119 older adults (≥ 55 years) enrolled in a University Program for Older Adults in southern Brazil (84.9% women; mean age: 68.7 ± 6.8 years). Body composition was assessed using bioelectrical impedance analysis, muscle strength by handgrip dynamometry, and physical performance by gait speed. Sarcopenia was defined according to EWGSOP2 criteria, and sarcopenic obesity was identified using different combinations of adiposity and muscle mass parameters. Metabolic, inflammatory, and oxidative stress markers were analyzed. Associations were evaluated using bivariate tests and Spearman correlations ($p \leq 0.05$). **Results:** The overall prevalence of sarcopenic obesity was 4.2% ($n = 5$), varying according to the diagnostic criteria applied. Women showed a higher prevalence of low skeletal muscle mass (65.3% vs. 35.3%) and reduced handgrip strength, while men had a higher risk of elevated body fat percentage (47.1% vs. 17.8%). Calf circumference was the only variable significantly associated with obesity ($p = 0.021$). Among women, adiposity indicators were positively correlated with triglyceride levels (BMI: $r = 0.263$, $p < 0.001$; fat mass: $r = 0.295$, $p < 0.001$), whereas gait speed was inversely associated with lipid peroxidation measured by TBARS ($r = -0.295$, $p < 0.001$), indicating lower oxidative damage with better functional performance. No significant sex differences were observed in glycemic, lipid, inflammatory, or oxidative stress markers ($p > 0.05$). **Conclusions:** Although the prevalence of sarcopenic obesity was low in this physically active older population, relevant associations were observed between adiposity, metabolic profile, oxidative stress, and functional performance. These findings highlight the importance of early screening of body composition, muscle strength, and redox imbalance, even in health-promoting settings, to prevent subclinical functional decline during aging.

Keywords: Sarcopenic Obesity; Oxidative Stress; Inflammation; Older Adults; Functional Capacity.

Introduction

Population aging is a global phenomenon that imposes important challenges to public health and is associated with a significant increase in the prevalence of geriatric syndromes that compromise functionality, autonomy and quality of life, with important clinical repercussions. Among these conditions, sarcopenia and obesity stand out, both of which are strongly related to metabolic, inflammatory, and functional changes throughout the course of aging [1].

Obesity is associated with a chronic low-grade inflammatory state and an increased risk of metabolic and cardiovascular diseases and insulin resistance. Sarcopenia is characterized by the progressive loss of muscle mass, strength and physical performance, and is related to the functional decline of the elderly [2]. The coexistence of these conditions configures sarcopenic obesity, a complex syndrome related to a higher risk of falls, hospitalizations, functional dependence and mortality [3].

From a pathophysiological point of view, sarcopenic obesity results from the interaction between chronic systemic inflammation, insulin resistance, hormonal changes, mitochondrial dysfunction and anabolic resistance, promoting a cycle of visceral fat accumulation and muscle loss that accelerates functional decline, even in individuals considered active [4]. In addition, traditional anthropometric indicators, such as body mass index, are insufficient to identify changes in body composition, requiring more specific assessments, such as bioelectrical impedance analysis (BIA) and functional tests [5].

Sarcopenia, alone or in association with obesity, is related to high personal, social, and economic burdens, in addition to a higher prevalence of cardiovascular and respiratory diseases, and cognitive impairment. In addition, it contributes significantly to reduced quality of life, loss of functional independence, need for long-term care, and increased mortality among the elderly [6]. In this sense, the early identification of sarcopenia, as well as its coexistence with obesity, even in apparently healthy and active elderly people, is a central element for the planning of preventive and comprehensive care strategies aimed at healthy aging.

Despite the growing scientific interest, studies investigating sarcopenic obesity in elderly participants in programs to promote active aging are still limited. Initiatives such as the Open University for the Third Age (UNATI) have shown a positive impact on several health outcomes; however, sarcopenic obesity may be present subclinically even in functionally active populations, which reinforces the importance of more detailed and continuous evaluations [7].

In this context, the present study aimed to investigate the relationship between sarcopenic obesity, inflammatory and redox profile and functional capacity in elderly participants in UNATI, aiming to contribute to early risk screening and the development of preventive strategies that promote healthy, functional aging with better quality of life.

Methods

This is a cross-sectional observational study, inserted in the context of the Senior Project (Health, Aging, Nutrition, Integration and Orientation), in which specific outcomes related to body composition, functional capacity and health biomarkers in people aged ≥ 55 years or older were analyzed. The research was carried out with participants of the Open University for the Third Age Program of the State University of Western Paraná, Francisco Beltrão campus (UNATI-FB), respecting the ethical guidelines for research with human beings and with the approval of the Research Ethics Committee (opinion no. 7,050,783; CAAE: 82364924.8.0000.0107).

The sample was composed of individuals aged ≥ 55 years, regularly enrolled in UNATI-FB activities, with sufficient mobility and communication skills to perform the proposed tests. Individuals who refused to participate, had advanced neoplastic diseases, did not attend the collection stages, or had sensory limitations that made the procedures unfeasible were excluded. All participants were previously informed about the objectives of the study and signed the Informed Consent Form. Data collection was carried out individually, by prior appointment, in Universidade Estadual do Oeste do Paraná (UNIOESTE) environments or in health units of the participating municipalities, by a previously trained multidisciplinary team.

Sociodemographic information was obtained using a structured questionnaire. Anthropometric assessment included weight, height, body mass index, waist circumference, and calf circumference. Body composition, including fat percentage and skeletal muscle mass, was assessed by BIA, following standardized protocols [8]. Skeletal muscle mass was estimated by a specific equation and adjusted for height, using the Janssen equation [9]. Functional capacity was assessed by means of gait speed in a standardized course [10,11], and handgrip strength, measured with a Saehan hydraulic dynamometer (model SH5001), considering specific cutoff points by sex and body mass index [11]. The level of physical activity was measured using the International Physical Activity Questionnaire (IPAQ), in its short version [27].

Sarcopenia was classified according to the criteria of the European Working Group on Sarcopenia in Older People (EWGSOP2), considering indicators of muscle mass, strength, and physical performance [12]. Sarcopenic

obesity was defined by the concomitant presence of sarcopenia and obesity, and different combinations of adiposity criteria (fat percentage, body mass index, and waist circumference) and muscle indicators were explored, as described in the literature [3].

Blood collection was performed after fasting following validated and standardized procedures, according to the Clinical and Laboratory Standards Institute [15], and the samples were processed and stored appropriately for subsequent analysis of biochemical, inflammatory, and oxidative stress biomarkers. The glycemic and lipid profile was determined by enzymatic colorimetric methods, according to the protocol described in the manufacturer's package insert [16]. The markers of oxidative stress were analyzed according to the protocol adapted by Lapenna *et al.* [17], and the antioxidant capacity were analyzed by standardized spectrophotometric techniques, following the method described by Vijayalakshmi *et al.* [18]. The immunological analysis was performed according to the protocol described by Engvall and Perlmann, using the enzyme-linked immunosorbent assay (ELISA) method [28].

Statistical analysis included descriptive statistics, with means and standard deviations for continuous variables and absolute and relative frequencies for categorical variables. The normality of the data was assessed using the Shapiro-Wilk test. Comparisons between genders were performed using the student's t-test or the Mann-Whitney U test, according to the distribution of the data. The associations between obesity, sarcopenia criteria, and metabolic, inflammatory, and oxidative profiles were analyzed using the chi-square test, with calculation of prevalence ratios and 95% confidence intervals, using Fisher's exact test when appropriate. Correlations were evaluated using Spearman's coefficient. A significance level of 5% was adopted, and the analyses were performed using SPSS version 21.0 [19].

Results

Initially, 187 elderly people linked to the Open University for the Third Age (UNATI) were interviewed. After exclusions due to the absence of laboratory data and/or handgrip strength ($n = 67$; 36.0%), the final sample eligible for analysis was composed of 119 participants (64.0%). The characteristics related to body composition, handgrip strength, and gait speed according to sex are presented in Table 1.

Table 1. Body composition, handgrip strength, and gait speed according to sex in older adults (≥ 55 years) enrolled in the Open University for the Third Age (UNATI), southern Brazil ($n = 119$), 2026.

Variable	Women ($n=101$) (Mean $\pm$ SD)	Men ($n=17$) (Mean $\pm$ SD)	<i>P</i>
Age (years)	68.20 $\pm$ 6.85	70.65 $\pm$ 6.44	0.172
Weight (kg)	70.21 $\pm$ 11.47	86.21 $\pm$ 15.34	<0.001
Height (m)	1.57 $\pm$ 0.06	1.69 $\pm$ 0.06	<0.001
BMI (kg/m ²)	28.31 $\pm$ 4.30	30.08 $\pm$ 4.44	0.121
MS (Kg)	19.03 $\pm$ 2.96	30.69 $\pm$ 5.74	<0.001
MS (%)*	27.43 $\pm$ 4.11	35.76 $\pm$ 4.42	<0.001
IMME/Est2 (kg/m2)*	7.68 $\pm$ 1.03	10.69 $\pm$ 1.61	<0.001
Massa magra (kg)	44.66 $\pm$ 6.12	60.37 $\pm$ 10.76	<0.001
Lean mass (%)	64.12 $\pm$ 5.18	70.17 $\pm$ 6.13	<0.001
Fat mass (kg)	25.52 $\pm$ 7.01	25.85 $\pm$ 7.74	0.863
Fat mass (%)	35.88 $\pm$ 5.18	29.84 $\pm$ 6.13	<0.001
Waist circumference (cm)	92.41 $\pm$ 10.77	104.41 $\pm$ 11.27	<0.001
Calf circumference (cm)	35.82 $\pm$ 3.71	37.08 $\pm$ 5.15	0.226
Handgrip (kgf)	27.00 $\pm$ 6.56	38.81 $\pm$ 6.91	<0.001
Walking (seconds)*	5.03 $\pm$ 1.20	4.65 $\pm$ 0.96	0.390
Walking (meters/sec)*	1.10 $\pm$ 0.26	1.02 $\pm$ 0.21	0.375

BMI: Body mass index (kg/m²); MS: Muscle mass (kg). Janssen MME = estimation of skeletal muscle mass by Janssen equation; MME percentage = proportion of skeletal muscle mass in relation to total body weight (%); IMME/Est² = height-adjusted skeletal muscle mass index (kg/m²), indicator of sarcopenia. OBS= Gait (woman $n=79$; man $n=14$); * Mann-Whitney U test.

Source: Elaboration by the authors (2026).

The sample was predominantly female ($n=101$; 84.9%), with a mean age similar to that of women and men ($p=0.172$). Men had significantly higher values of body weight, height, absolute and relative skeletal muscle mass, height-adjusted muscle mass index, lean mass, and handgrip strength ($p<0.001$). Women had higher body fat percentage ($p<0.001$) and lower muscle strength values, characterizing important differences in body composition between the sexes. Despite this, no statistically significant differences were observed between women and men in terms of body mass index and gait speed, indicating preserved functional performance in most of the sample. Waist circumference showed high mean values, according to recommendations in the literature, in both sexes, with

significantly higher averages among men ($p<0.001$), suggesting a high prevalence of central obesity in the population studied. Serum glycemic, lipid, inflammatory, and oxidative stress markers according to sex are presented in Table 2.

Table 2. Serum concentrations of glycemic and lipid profile, inflammatory and oxidative stress markers according to sex.

Variable	Women (n=101) (Mean $\pm$ SD)	Men (n=17) (Mean $\pm$ SD)	P
Glucose (mg/dL)	105.6 $\pm$ 30.15	98.62 $\pm$ 16.51	0.631
Total cholesterol (mg/dL)	182.01 $\pm$ 41.57	191.48 $\pm$ 39.29	0.359
Triglycerides (mg/dL)	124.33 $\pm$ 43.58	111.23 $\pm$ 31.91	0.308
TNF α	103.86 $\pm$ 139.37	121.43 $\pm$ 122.74	0.467
Ferricyanide (ug/mL)	1036.41 $\pm$ 297.24	1085.69 $\pm$ 225.90	0.563
TBARS (nmol/mL)	25.05 $\pm$ 9.51	24.63 $\pm$ 12.63	0.304
Vitamin C (mg/dL)	7.72 $\pm$ 6.32	6.87 $\pm$ 0.49	0.287
IL6 (pg/mL)	101.23 $\pm$ 232.69	85.91 $\pm$ 161.19	0.830
P-SH (nmol/mg)	4.42 $\pm$ 2.57	3.72 $\pm$ 1.45	0.372
NP- SH (GSH) (μ mol/L)	5.47 $\pm$ 2.74	5.85 $\pm$ 2.55	0.229

TNF α : Tumor necrosis factor alpha; **TBARS:** Thiobarbituric acid reactive substances; **IL-6:** Interleukin-6; **P-SH:** Protein sulfhydryl groups; **NP-SH:** Non-protein sulfhydryl groups; **GSH:** Reduced glutathione. * For glycemic and lipid profile variables, n=98 for women; for TNF, n= 11 for men and n=78 for women; for IL6, n= 79 for women and n= 10 for men.

Source: Elaboration by the authors (2026).

No statistically significant differences were observed between women and men for glucose, total cholesterol, triglycerides, TNF- α , IL-6, TBARS, vitamin C, and thiol markers ($p>0.05$). Mean glucose concentrations were close to or above the reference values in both sexes, indicating metabolic impact associated with aging, regardless of gender. Inflammatory and oxidative stress markers showed variability, but no significant differences between men and women. Characteristics related to obesity, sarcopenia, and physical activity according to sex are presented in Table 3.

Table 3. Characteristics related to obesity, sarcopenia, and physical activity according to sex.

Variable	Women (n=101)		Men (n=17)	
	n	%	n	%
% fat (BIA)				
Suitable	83	82.2	9	52.9
Risk	18	17.8	8	47.1
Body Mass Index (BMI)				
Suitable	42	41.6	6	35.3
Risk	59	58.4	11	64.7
Low skeletal muscle mass (SMM)				
No	35	34.7	11	64.7
Yes	66	65.3	6	35.3
Calf circumference				
Suitable	77	76.2	10	58.8
Risk	24	23.8	7	41.2
Handgrip strength				
Suitable	76	75.2	15	88.2
Low	25	24.8	2	11.8
Adjusted Handgrip Strength (BMI)				
Suitable	90	89.1	15	88.2
Low	11	10.9	2	11.8
Gait Speed				
Adequate	72	91.1	13	92.9
Reduced	7	8.9	1	7.1
Physical activity level (IPAQ)				
Very active	9	8.9	1	5.9
Active	46	45.5	7	41.2
Irregularly active A	27	26.7	5	29.4
Irregularly active B	16	15.8	3	17.6
Sedentary	3	3.0	1	5.9

OBS= Gait (woman n=79; man n=14); IPAQ = International Physical Activity Questionnaire
Source: Authors' elaboration (2026).

Table 3 shows that low skeletal muscle mass was more prevalent among women (65.3%) compared to men (35.3%). On the other hand, the risk by body fat percentage was higher among men (47.1% vs. 17.8%). Reduced handgrip strength values were more frequent in females, both in the analysis without adjustment and after adjustment for BMI, suggesting greater functional vulnerability among women. Despite this, most participants had adequate gait speed. Regarding the level of physical activity, there was a predominance of individuals classified as active or irregularly active A in both sexes, with low percentages of sedentary lifestyle. The associations between obesity and sarcopenia criteria and the metabolic profile are presented in Table 4.

Table 4: Association between obesity and sarcopenia criteria and metabolic profile (N=115).

Variables	Obesity		p	RP (95% CI)
	Yes n (%)	No n (%)		
Handgrip strength				
Adequate	22 (84.6)	69 (75.0)	0.429*	0.545 (0.17-1.75)
Risk	4 (15.4)	23 (25.0)		
Strength Adjusted Palm Pressure BMI				
Adequate	23 (88.5)	82 (89.1)	1.000*	1.07 (0.27-4.21)
Risk	3 (11.5)	10 (10.9)		
Skeletal muscle mass				
Adequate	11 (42.3)	35 (38.0)	0.694*	0.84 (0.35–2.03)
Risk	15 (57.7)	57 (62.0)		
Calf circumference				
Adequate	24 (92.3)	63 (68.5)	0.021*	0.181 (0.04-8.82)
Risk	2 (7.7)	29 (31.5)		
Gait Speed				
Adequate	66 (89.2)	19 (100.0)	0.200*	0.89 (0.82-0.97)
Reduced	8 (10.8)	0 (0.0)		
Glucose				
Adequate	14 (58.3)	42 (46.2)	0.288**	1.63 (0.66-4.06)
Up	10 (41.7)	49 (53.8)		
Total cholesterol				
Adequate	14 (58.3)	53 (58.2)	0.994**	1.00 (0.40-2.50)
Up	10 (41.7)	38 (41.8)		
Triglycerides				
Adequate	20 (83.2)	66 (72.5)	0.086*	0.50 (0.20-1.49)
Up	4 (16.7)	25 (27.5)		

*Fischer's exact test. ** Chi-square test.

Source: Authors' elaboration (2026).

Table 4 shows the association between obesity and sarcopenia criteria, functional performance, and metabolic profile. In the bivariate analysis, no statistically significant associations were observed between obesity and handgrip strength, BMI-adjusted strength, skeletal muscle mass, gait speed, blood glucose, total cholesterol, and triglycerides ($p>0.05$). Calf circumference was the only variable that showed a statistically significant association, indicating a lower prevalence of risk for low calf circumference among obese individuals ($p=0.021$), although with a wide confidence interval.

The prevalence of sarcopenic obesity according to different evaluation criteria is presented in Table 5.

Table 5: Prevalence of sarcopenic obesity according to different evaluation criteria.

	Women (n=101)		Men (n=17)	
	n	%	n	%
Sarcopenic obesity				
%gordura (BIA) + sarco. (MS + PP)				
No	101	100	16	94.1
Yes	0	0	1	5.9
%fat (BIA) + sarco. (CP+PP)				
No	101	100	16	94.1
Yes	0	0	1	5.9
Gord. central (CC) + sarco. (MS + PP)				
No	97	96.0	17	100
Yes	4	4.0	0	0
Obesity (BMI) + sarco. (CP+PP)				
No	101	100	17	100
Yes	0	0	0	0

BIA: Bioelectrical impedance analysis; **% fat:** Body fat percentage; **MS:** Muscle mass; **PP:** Handgrip strength; **CP:** Calf circumference; **CC:** Waist circumference; **BMI:** Body mass index; **Sarco.:** Sarcopenia.
Source: Elaboration by the authors (2026).

Table 5 shows the prevalence of sarcopenic obesity, and it is observed that there was variation according to the criteria adopted. Among women, the condition was identified only when the combination of central fat and sarcopenia based on skeletal muscle mass (SMM) and handgrip strength (PP) (4.0%) was used. Among men, only one case (5.9%) was identified when the combined criterion of body fat percentage and sarcopenia was used. Considering all criteria, the overall prevalence of sarcopenic obesity was 4.2% in the total sample.

The correlations between obesity, sarcopenia, and serum glycemic and lipid profile, inflammatory, and redox markers are presented in Table 6.

Table 6. Correlation between obesity, sarcopenia and serum glycemic and lipid profile, inflammatory and redox markers.

	Glucose	Total Cholesterol	Tiglycerides	TNF α	Ferricianide	TBARS	Vitamin C	IL6	P-SH	NP-(GSH)	SH
Women (n=98)											
BMI (kg/m ²)	0.190	-0.068	0.263**	0.096	0.198*	0.106	0.048	0.123	0.010	0.017	
MS (Kg)	0.210*	0.019	0.034	0.138	0.020	0.006	-0.041	0.143	-0.004	-0.152	
Fat mass (kg)	0.041	-0.096	0.295**	0.110	0.199*	0.92	0.022	0.117	0.013	0.089	
Fat mass (%)	-0.042	-0.11	0.248*	0.008	0.192	0.070	0.039	0.017	-0.002	0.168	
Circ. Pantu. (cm)	0.034	0.018	0.143	0.179	0.037	0.126	-0.30	0.255*	-0.022	0.017	
Preen. pal. (kgf)	0.153	0.043	0.034	0.038	0.077	0.228*	-0.131	0.044	0.013	-0.132	
Gait (m/s)	-0.086	-0.267*	0.160	0.028	-0.128	-0.295**	-0.113	0.161	-0.122	0.023	
Men (n=17)											
BMI (kg/m ²)	0.257	-0.015	0.127	-0.670*	< 0.001	0.088	-0.193	-0.537	-0.006	0.103	
MS (Kg)	0.375	0.223	-0.115	-0.628*	0.120	0.098	0.150	-0.330	-0.087	0.132	
Fat mass (kg)	-0.086	-0.145	0.201	0.716*	-0.066	0.299	-0.178	-0.213	0.365	0.279	
Fat mass (%)	-0.348	-0.145	0.267	-0.623*	-0.025	0.137	-0.183	-0.330	0.264	0.333	
Circ. Pantu. (cm)	0.011	-0.034	0.149	-0.542	-0.034	0.038	-0.172	-0.449	-0.194	0.214	
Preen. pal. (kgf)	-0.338	0.502*	-0.220	-0.152	-0.087	0.086	0.047	0.394	0.011	0.132	
Gait (m/s)	0.204	-0.222	0.095	-0.441	-0.042	-0.152	-0.253	-0.533	-0.059	0.371	

*P<0,05; **p<0,001; OBS= TNF α (women n = 76; men n = 11); IL-6 (women n = 77; men n = 10)

Source: Elaboration by the authors (2026).

Table 6 shows weak to moderate correlations between obesity indicators, body composition, muscle strength, and functional performance with markers of glycemic, lipid, inflammatory, and oxidative stress profiles, with distinct patterns between the sexes. Among women, body mass index and fat mass, absolute and percentage, were positively correlated with triglycerides, while gait speed showed a negative correlation with total cholesterol and TBARS, indicating an association between better functional performance and lower oxidative stress. In men, inverse correlations were observed between body mass index and skeletal muscle mass with TNF- α , as well as a positive correlation between fat mass and this inflammatory marker, in addition to a positive association between handgrip strength and total cholesterol, evidencing the complexity of the interactions between adiposity, muscle function, and metabolic and inflammatory processes. The individual characteristics of sarcopenic obese patients are presented in Table 7.

Table 7. Individual description of sarcopenic obese patients (N=5).

	Subject 1	Subject 2	Subject 3	Subject 5	Subject 6
Continuous data	Men	Women	Women	Women	Women
Age (years)	74	69	76	77	71
BMI (kg/m ²)	26.8	30.1	31.2	30.2	29.7
MS (Kg)	22.42	17.87	19.95	17.41	18.94
MS (%)*	34.28	26.76	26.60	26.63	27.65
Fat-free mass (kg)	44.1	41.6	46.1	40.4	43.1
Fat-free mass (%)	67.4	62.3	61.5	61.5	63.4
Fat mass (kg)	21.3	25.2	28.9	25.0	24.9
Fat mass (%)	32.6	37.7	38.5	38.2	36.6
Waist circumference (cm)	97.0	94.0	96.0	98.0	101.0
Calf circumference(cm)	34.3	35.0	44.5	35.5	37.0
Handgrip (kgf)	27.5	15.0	16.0	17.0	20.0
Walking (seconds)	--	5.98	6.31	7.76	4.40
Gait (meters/sec)	--	1.31	1.38	1.70	0.96
Glucose (mg/dL)	83.32	89.66	103.77	81.5	132.90
Total cholesterol (mg/dL)	149.81	155.94	199.10	146.20	156.89
Triglycerides (mg/dL)	120.34	114.65	88.60	108.74	146.17
TNF α	--	14.30	13.99	478.56	13.99
Ferrician (ug/mL)	984.33	700.09	945.73	1291.39	1280.86
TBARS (nmol/mL)	14.36	17.99	15.85	27.85	43.71
Vitamin C	6.54	7.29	6.86	7.78	8.89
IL6	--	0.49	13.16	23.27	136.68
P-SH	2.42	0.40	5.60	7.00	3.47
NP-SH (GSH) (μ mol/L)	4.98	5.84	3.58	2.08	5.21
Categorical Data					
% fat (BIA)	Risk	Suitable	Suitable	Suitable	Suitable
BMI	Overweight	Overweight	Overweight	Overweight	Overweight
Waist circumference	Suitable	Risk	Risk	Risk	Risk
Sarcopenia (PP and MME)	Yes	Yes	Yes	Yes	Yes
Sarcopenia (PP and Calf)	Yes	No	No	No	No
Skeletal Muscle Mass	Risk	Risk	Risk	Risk	Risk
% Skeletal Muscle Mass	Risk	Risk	Risk	Risk	Suitable
Calf circumference	Risk	Suitable	Suitable	Suitable	Suitable
Handgrip strength	Low	Low	Low	Low	Low

Adjusted PP Strength - BMI	Low	Low	Low	Low	Low
Gait Speed	--	Adequate	Adequate	Adequate	Adequate
Physical activity level	Irregularly active B	Active	Sedentary	Active	I. Asset B
Glucose (mg/dL)	Normal	Normal	Up	Normal	Up
Total cholesterol (mg/dL)	Normal	Normal	Up	Normal	Normal
Triglycerides (mg/dL)	Normal	Normal	Normal	Normal	Normal
%gor (BIA) + sarco. (MS + PP)	Yes	No	No	No	No
%gor (BIA) + sarco. (CP+PP)	Yes	No	No	No	No
Gord. (CC) + sarco. (MS + PP)	No	Yes	Yes	Yes	Yes
Obesity (BMI) + sarco. (CP+PP)	No	No	No	No	No

BMI indicates body mass index; MME refers to skeletal muscle mass; MME (%) corresponds to the percentage of skeletal muscle mass in relation to total body weight; PP denotes handgrip strength; NC indicates calf circumference; CC refers to waist circumference (abdominal); BIA stands for bioelectrical impedance analysis; TNF- α corresponds to tumor necrosis factor alpha; IL-6 denotes interleukin-6; TBARS refers to thiobarbituric acid reactive substances, a marker of lipid peroxidation; P-SH represents protein thiol groups; and NP-SH (GSH) denotes non-protein thiols (reduced glutathione).

Source: Authors' elaboration (2026).

Table 7 describes the five participants classified as sarcopenic obese, who are predominantly women, aged 69 to 77 years, all overweight, with reduced skeletal muscle mass, and decreased handgrip strength. Despite the preservation of gait speed in most cases, a high percentage of body fat and waist circumference of risk was observed, in addition to variability in the level of physical activity. The metabolic profile showed punctual alterations in blood glucose and total cholesterol, while inflammatory and oxidative stress markers showed wide variability, reinforcing the heterogeneous and multifactorial character of sarcopenic obesity even in functionally active older adults.

Discussion

The present study analyzed the association among sarcopenic obesity, inflammatory profile and functional capacity in the elderly at UNATI, showing significant differences in body composition between the sexes, although functional performance was preserved in most of the sample. Men had higher muscle mass and handgrip strength, while women had higher body fat percentage and higher prevalence of low muscle mass. These results are in line with the literature, which describes higher adiposity and lower muscle mass and strength in women, and higher muscle mass associated with a higher prevalence of central obesity in men [6,20], differences attributed to biological and hormonal factors, and to the distinct patterns of body distribution in aging, configuring a profile that favors greater relative adiposity in women and greater amount of fat-free mass in men, although both may present cardiometabolic risk related to the accumulation of abdominal fat [22].

Even with the maintenance of gait speed and adequate levels of physical activity, a high prevalence of low muscle mass and reduced strength was observed, especially among women, suggesting a subclinical risk for sarcopenia. As highlighted by Pontes [23], changes in muscle mass and strength may precede evident functional manifestations, configuring an initial stage of vulnerability not yet detected by functional tests, which reinforces the importance of early screening to prevent progression to established sarcopenia and more pronounced functional decline.

In the present study, glycemic, lipid, inflammatory, and oxidative stress markers did not show significant differences between the sexes. This absence of differences should be interpreted considering the high interindividual variability typical of aging, in addition to the small sample size among men and the smaller number of available cytokine samples, which limits statistical power [24, 25]. Still, the correlation patterns suggest biologically plausible relationships: among women, adiposity indicators correlated with triglycerides, and better gait performance was associated with lower levels of TBARS, suggesting that functional status may be related to lower oxidative stress. Among men, associations between adiposity, muscle mass, and TNF- α have been observed, although these findings should be interpreted with caution due to the small number of observations [26,27,28].

An important result was the low prevalence of sarcopenic obesity (4.2%), with important variation according to the criteria adopted for adiposity and sarcopenia. This result reinforces the limitation of comparability among studies and the impact of the operational definition on the estimation of prevalence, especially in samples with good functional performance. Furthermore, calf circumference was the only variable associated with obesity, with a lower frequency of values classified as low among obese individuals. This result may reflect limitations of the indicator in individuals with higher total BMI, since circumference can capture non-muscle components and not adequately discriminate muscle quality, which reinforces the need to interpret anthropometric measurements in conjunction with the assessment of strength and body composition [12].

The findings reinforce the importance of screening strategies in active aging programs that go beyond apparent functional assessment, incorporating detailed analyses of body composition, muscle strength, and metabolic markers, in order to identify conditions such as sarcopenia and sarcopenic obesity early. Among the limitations, the cross-sectional design, the small number of men, sample losses (with potential selection bias), the use of bioelectrical impedance, and the low prevalence of the condition stand out, restricting the generalization of findings and hindering stratified analyses. Longitudinal studies, with larger samples, are needed to deepen the understanding of the relationships between sarcopenic obesity, inflammation, and decline in strength and functionality in aging.

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Conflict of interest

The authors declare that there is no conflict of interest related to this study.

Statement of Ethical Guidelines

The study was approved by the Research Ethics Committee, under opinion No. 7.050.783 (CAAE: 82364924.8.0000.0107), ensuring respect for autonomy, confidentiality of information, and obtaining the Informed Consent Form from the participants.

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ANEXOS

Normas da revista científica: Journal of Cachexia, Sarcopenia and Muscle (WILEY).



Manuscript Categories and Criteria

The Journal of Cachexia, Sarcopenia and Muscle accepts the following categories of articles:

Original Articles

These should not exceed 4500 words (excluding abstract, references, tables and figures) and may include up to 8 figures (max of 6 sub-sections per figure) and/or tables and up to 40 references. Additional figures, tables and references should be submitted as supplement.

The article should be divided into the following sections: Title page, Abstract, Keywords, Introduction, Methods, Results, Discussion, Acknowledgements (funding, conflict of interest, ethical guidelines statement).

The abstract of Original Articles should be divided into the sections Background, Methods, Results, Conclusions and should not exceed 400 words. Important note: the abstract needs to be data rich and contain all main study results. Descriptive results abstracts are not acceptable. For clinical trial submissions, see extra comments.

Reporting on Clinical Trials

- Typically results should be presented according to intention-to-treat (for efficacy). Reporting on safety data and trials using non-inferiority designs may use other methodology.
- Biostatistical approaches should be described and the responsible biostatistician should be identified. Note: a clinical trial can only have one primary efficacy endpoint, unless alpha-splitting or a Hochberg procedure (or similar approaches) are used. Please submit a copy of the statistical analysis plan in English
- Submit the trial protocol in an English translation as appendix.
- Consider working the trial statistician together with an experienced trialist to bring the reporting of the study to highest standards.
- The trial must have been registered before its start. Please provide registration details (i.e. the NCT number etc).

Comprovante de submissão do artigo científico

Journal of Cachexia, Sarcopenia and Muscle	
Original Article	
IMPACT OF SARCOPENIC OBESITY ON OXIDATIVE STRESS, INFLAMMATORY FACTORS, AND FUNCTIONAL CAPACITY IN OLDER ADULTS	
Submission ID	c508a534-06a0-475a-ba33-dcdb06b983a8
Manuscript ID	8082965
Submission Version	Initial Submission
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Parecer do Comitê de Ética em Pesquisa

PARECER CONSUBSTANCIADO DO CEP

DADOS DO PROJETO DE PESQUISA

Título da Pesquisa: PROJETO SENIOR (Saúde, Envelhecimento, Nutrição, Integração e Orientação)

Pesquisador: Lirane Elize Defante Ferreto

Área Temática:

Versão: 1

CAAE: 82364924.8.0000.0107

Instituição Proponente: UNIVERSIDADE ESTADUAL DO OESTE DO PARANÁ

Patrocinador Principal: FUNDO PARANÁ
Financiamento Próprio

DADOS DO PARECER

Número do Parecer: 7.050.783

Apresentação do Projeto:

O projeto Senior irá investigar 160 participantes da Universidade da Terceira Idade (UNATI) utilizando o método de Avaliação Geriátrica Compreensiva (AGA) para diagnosticar e monitorar a saúde de idosos, focalizando nutrição e saúde. Trata-se de um projeto institucional da Unioeste.

Objetivo da Pesquisa:

Segundo PB (p.03), são objetivos: *Objetivo Primário:

Aplicar a Avaliação Geriátrica Ampla (AGA) como ferramenta para monitorar, avaliar e melhorar a saúde e o bem-estar de idosos, identificando

necessidades específicas e promovendo intervenções personalizadas que visem a manter a autonomia, prevenir doenças e melhorar a qualidade de vida dessa população.

Objetivo Secundário:

a)Avaliar as capacidades funcionais e cognitivas dos idosos por meio de testes padronizados.b)Realizar avaliações para identificar condições

crônicas prevalentes;c)Avaliar o estado nutricional dos idosos e propor intervenções dietéticas para prevenir desnutrição.d)Analisar as redes de

suporte social e as condições psicossociais dos idosos.e)Identificar riscos de quedas e

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